

Fizzim: The Finite State Machine Design Tool

Fizzim is a GUI-based, independent Finite State Machine (FSM) designing tool that offers flexibility of modifying the Verilog output without touching the GUI. Read this article to know how to start working with it

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Finite State Machines (FSMs) are frequently used in digital designs and FSM designing has become a common task for ASIC design engineers. FSMs can be coded directly into Verilog but one can always design them in a GUI and allow software to generate the Verilog code. Many GUI-based environments have been developed but no commercial tool has really been able to achieve widespread acceptance.

Fizzim is an open source graphical FSM design tool, the GUI for which is written in Java for portability and its back-end code generation written in Perl. Fizzim has its GUI and Verilog output independent of each other, with virtually everything being implemented as attributes, which allows for new backend (Verilog generation) features to be added without touching the GUI.

Starting Fizzim

Java-based GUI makes Fizzim platform independent. It is distributed as a '.jar' (Java archive) file and you can simply run it using Sun Java Runtime Environment.

You can open the executable jar file with the Java Runtime Environment or open terminal and run it using the following command line:

```
java -jar fizzim_v11.09.25.jar
```

You can also add the fizzim file on the command line:

```
java -jar fizzim_v11.09.25.
```

```
jarfilename.fzm
```



Fig. 1

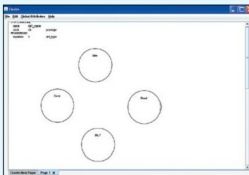


Fig. 2

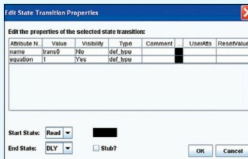


Fig. 3

System requirements

Runs on Windows, Linux, Mac OS
Note: Installation file can be found in EFY Plus DVD that came with this magazine.

GUI

The easy-to-learn GUI offers intuitive options. You can create new states and transitions by right clicking in open space and selecting the option from the menu. Right clicking on an object gives you a menu to edit the object whilst double (left) clicking on an object will bring up the properties menu for that object. Select *Edit -> undo* or *ctrl+Z* to undo and *Edit -> redo* or *ctrl+Y* to redo. There is no limit on undo/redo.

Creating new states. Let us take a simple FSM example, i.e., Cliff Cummings basic FSM to create a basic state machine using Fizzim. Right clicking and selecting *New State* opens a new menu as shown in Fig. 1. Change the state name as *Idle* and click *OK*. Repeat the same steps to create three more states as in Fig. 2. You can drag the states by left clicking and holding onto a particular state.

Creating transitions. To create a new transition, either right click on the start state and select *Add State Transition* to and the desired end state, or right click on the open space and select *New State Transition*, which opens a full menu as shown in Fig. 3. Select the start and end states in the menu. Perform these steps to have all the transitions as shown in Fig. 4.